

ALGORITHMIC TRADING EFFECT ON MARKET LIQUIDITY DURING A FLASH CRASH: EVIDENCE FROM THE INDIAN MARKET

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ABSTRACT

The study contributes to the emerging literature of algorithmic trading (AT) and flash crash by investigating the impact algorithmic traders have on the Indian National Stock Exchange (NSE) during the October 5th 2012 flash crash. The study examines how the flash crash affected market liquidity, how the AT intensity changes during flash crash as well as the role of AT and non-AT impacting market liquidity during and after flash crash period. We utilize flash crash event day with normal trading days and time in our investigation. We find 91.1 percent of order submission through AT around the flash crash event. Both graphic representation, univariate testing and multivariate regressions show that market liquidity and AT-intensity decreases during flash crash time, AT affects market liquidity negatively during flash crash time and non-AT provides more liquidity to the market during normal trading day. Our results are consistent after controlling for the firm specific effects.

Keywords: Algorithmic trading, Orderbook, flash crash, liquidity.

1. INTRODUCTION

Improvements in trading facilities on exchanges enhance trading mechanism, price discovery process and reduce trade risk. The advancement in technologies and market structure innovations in last two decades has dramatically improved the market quality, liquidity and trade execution speed. Most of the exchanges around the world allow high frequency algorithmic trading. The growth in high-frequency trading has attracted attention from academicians, practitioners, and market regulators. The recent studies suggest that high frequency trading is more popular in the equity and futures market¹ (Zhang, 2010; O'Reilly, 2012; Easley et al., 2012; Scholtus et al., 2014). The reliance on high frequency trading with new technologies has also raised a serious concern related to price manipulation (Biais and Woolley, 2011; Aitken et al., 2015), quote stuffing (Egginton et al, 2016), price volatility (Hendershott and Riordan, 2013; Subrahmanyam; 2013) and impact on market quality (Brogaard, 2010; Hendershott et al., 2011; Easley et al., 2012, Jones, 2013; Biais et al, 2015; Foucault et al, 2015). Many exchanges have witnessed episodes of flash crashes after the introduction of high frequency trading facilities.² Among them are flash crashes on May 06, 2010 on New York stock exchange (NYSE) in the US, October 05, 2012 on National Stock exchange (NSE) and April 23, 2013 on Singapore stock exchange (SSE). The frequent occurrence of flash crash has raised a serious question on role of algorithmic trading (AT) on speed of order execution and overall market quality (Hendershott and Riordan; 2013). The AT software

¹ It accounts around 50–70% of equity trades and 50% of the futures market in the U.S., 40% in Canada, and 35% in London.

² Article published in Automated Trader Magazine Issue 39 Q2 2016. “Regulation AT: What investment managers need to know”. <http://www.automatedtrader.net/articles/compliance/155464/regulation-at-what-investment-managers-need-to-know>.

logic reacts very fast for a submitted order. Therefore, even a small wrong order may create a severe panic and cause a crash due to AT.^{3,4}

This study examines the effect of algorithmic trading activity on market liquidity during the flash crash on the NSE which happened on October 05, 2012. We analyze how algorithmic trading channel may exaggerate market panic which was due to human error and may cause a flash crash.⁵ Our study also compares flash crash event day with normal trading days to understand role of algorithmic trading on overall market liquidity in consideration of the flash crash event. We contribute to the line of research that examines the relation between AT and market liquidity by considering the order submission and its impact on overall market liquidity. The existing papers in the AT literature distinguish among the impacts of diverse algorithmic trading (AT) participants. Our study is in line with Hagstromer and Norden (2013), Biais and Foucault (2014), and others in addressing this shortcoming and demonstrating its importance to understanding the role of AT in financial markets in particular during the sudden downturns.

We find that quoted spread increases and market depth decreases during flash crash time, compared to normal trading time. This indicates, the market liquidity decreases during flash crash time. During flash crash time we observe a significant reduction in AT-intensity. We also find that AT affects market liquidity negatively during flash crash and non-AT provides more liquidity to the market during flash crash time.,

³ News Article published on October 26, 2015 in the Economic Times. “Is high frequency trading a necessary evil?”, <http://blogs.economictimes.indiatimes.com/et-commentary/is-high-frequency-trading-a-necessary-evil/>.

⁴ Article published regarding concern of SEBI on algo-trading. “Sebi asks stock exchanges to strengthen system audits”, <http://www.livemint.com/Money/ScYH4zH1HK9z0iUN2xNmrL/Sebi-asks-stock-exchanges-to-strengthen-system-audits.html>,

⁵ Freak trade at NSE: Rs 3 lakh cr erased in 2 minutes, http://www.business-standard.com/article/markets/freak-trade-at-nse-rs-3-lakh-cr-erased-in-2-minutes-112100600062_1.html

We contribute to the literature in several ways. First, we provide new evidence about flash crash events by examining a flash crash that has never been covered in the literature. Flash Crash papers have mostly been conducted on the U.S event (see Easley et al., 2012; Rose, 2011; Barrales, 2012; Boulton et al., 2014; McInish et al., 2014) by looking at an Indian stock market's Flash Crash we add a new perspective to this developing field of research. Second, to the best of our knowledge, this is the first study investigating impact of algorithmic trading on the intensity of the flash crash in Indian market. Thus our study contributes significantly to expand the understanding of algorithmic trading behavior in market stress time in emerging markets. It provides new insight into the behavior of AT and non-AT trading during flash crash time and in normal market time, which would enable a deeper understanding of the market structure and behavior. We also contribute to on-going debate on the role of algorithmic trading on order flow and trade execution. We use a unique orderbook dataset that enable us to separate orders into Algo, Non-algo and normal retail orders.

Our study is organized as follows. Section 2 gives some background on the flash crash on NSE in India. Section 3 presents a brief review of the related literature and proposes testable hypotheses. Section 4 discuss data and methodology. Section 5 describes results and findings. Last section provides our conclusion.

2. ALGORITHMIC TRADING AND FLASH CRASH IN INDIA

High frequency Algorithmic Trading (AT) in India was launched by Credit Suisse's Advanced Execution Services (AES) on June 22, 2009. Initially it was only focused around the Indian equities. The AT has started gaining popularity among traders and brokers in June 2010 when NSE

has allowed co-location facilities for the trading.⁶ Co-location allows broker member servers' to be placed side by side to the exchange server to reduce latency. It reduced the time taken in the transmission of data (orders) from broker terminals to exchange servers for the successful trade execution. Since speed in order submission is the key for successful trading, most of the brokerage firms have setup their server terminals closed to the exchange servers.

On October 5, 2012 the National Stock Exchange (NSE) was forced to halt trading briefly after 59 erroneous orders that caused the Nifty index to drop 960 points as much as 15.6 per cent within minutes. This event led a halt for about 15 minutes. The later investigation by the NSE has disclosed that the flash crash occurred due to human error. Emkay Global, a brokerage firm, has entered erroneous orders of worth \$125.7mn on behalf of it one of the institutional investor.⁷ The exchange regulatory body, Securities and Exchange Board of India (SEBI), has raised several questions on NSE's risk management tools and ordered for setting up more robust trading systems.⁸

3. LITERATURE REVIEW AND HYPOTHESES

The 1990's saw the creation of the limit order, in the 2000's algorithmic trading began, and over the past 10 years exchanges around the world have been adopting co-location facilities inside their exchanges. Co-location has resulted in a vast increase in trading volume, almost all due to the sharp increase in algorithmic trading. The drastic change in our markets caused by AT has led to questions about the effects it has on market liquidity and non-AT investors. The

⁶ Co-location facility at exchange premises allow to achieve lower latencies and faster execution.

⁷ Indian shares suffer \$60bn 'flash crash', <https://www.ft.com/content/9869b272-0ecd-11e2-9343-00144feabdc0?mhq5j=e1>

⁸ Emkay flash crash: Sebi censures NSE, http://www.business-standard.com/article/markets/emkay-flash-crash-sebi-censures-nse-114101001365_1.html

interest garnered by AT (especially high frequency trading) has not been examined extensively. Many questions remain unanswered.

Most of literature has reached similar conclusions; AT fundamentally improves liquidity, reduces volatility and plays an important role in price discovery. However, some argue that AT, and HFT especially, can cause volatility spikes, increase chances of systemic risk, consuming only cheap liquidity and profiting on the behalf of regular non-at investors (Jain et al., 2016). Hendershott & Moulton (2011) find that following the adoption of the New York Stock Exchanges hybrid system (which reduced execution time from ten seconds to less than one) resulted in a 10% increase in the effective spread along and increase in adverse selection cost. Hendershott & Riordan (2013) present the role of AT in the market and the how they react to changing spreads and market conditions. They find that although AT consumes cheap liquidity and sells when it is expensive, AT plays an important role in price discovery. Kelejian & Mukerji (2016) studied statistical arbitrage, proving that stock price volatilities (much of the time caused by high frequency trading) can significantly spillover to other industries and that price impact from AT is much higher than non-AT investors.

Commonly, algorithms are programmed to trade based off; quoted spreads (Hendershott & Riordan, 2013), macro-economic new announcements (Scholtus, Dijk & Frijns, 2014) and correlations with other company stocks (O'Hara, 2014). Strategies used by both AT and human traders may be similar in nature but processed at a speed faster than your eye can blink. Algorithmic traders can handle a higher quantity of information and process it in a fraction of a time than a human (Kelejian and Mukerji, 2016). This difference allows AT's to trade faster, resulting in profits based off speed advantage. Although a non-AT investor may have intuition and a larger set of parameters, the advantage of speed has led to a vast increase in AT trading

volume over the past five years on markets across the world. High frequency trading (large subset of AT) accounts for 60% of equity trades and 50% of future trading in the United States, 40% of equity traders in Canada and 35% in London (Aitken, Cumming & Zhang 2015).

Literature surrounding Flash Crashes has only focused on the one event in the U.S. Many studies specifically focus on the direct causes that triggered the event. Menkveld & Yueshen (2013) displayed a broad anatomy of the Flash Crash, finding that one sell order caused a significant long-run price impact during the Flash Crash time. Although the Flash Crash did not have any long-term welfare effects, the event may have increased implied volatility spreads and triggered a loss in investor confidence in the market, Kumiega, Sterijevski, & Vliet (2016). Kirilenko, Kyle, Samadi & Tuzun (2017) investigate the role and behavior of HFT's during the Flash Crash. The research finds that HFT's trade in the direction of price changes and comprised a significant portion of trading volume. Furthermore, competition between HFT's for liquidity simultaneously exacerbated price volatility. Although this study recognizes the benefits of HFT and improved technology in the market, "appropriate safeguards must be implemented to keep pace with trading practices" Kirilenko et al. (2017).

Our study uses the behavior characteristics Kirilenko, Kyle, Samadi & Tuzun (2017) had established. We use a normal trading day as a comparison to outline the effects of AT. The first hypothesis examines the aggregate effects of the Flash Crash upon market liquidity. Our liquidity measure includes two variables: quoted spread and market depth.

H1: Market liquidity measured as quoted spread and market depth significantly reduces during a flash crash compared to a normal trading time

Recently there is an attempt to find the impact AT, and specifically high frequency trading has on the market and if the advantages exceed the risk that new technology impose. Albeit different

conclusions, much of the research suggests that AT fundamentally improves liquidity, reduces volatility and plays an important role in price discovery. Aggarwal & Thomas (2013) analyze AT using India's National Stock Exchange and found that increased AT intensity has improved liquidity and reduced volatility. Brogaard, Hendershott & Riordan (2014) look directly at the impact AT has on price discovery and found that AT contributed to the process. Jovanovic & Menvelo (2011) present that HFT are liquidity suppliers and update quotes immediately after the arrival of news, which ultimately contributed to reduce adverse selection cost. Viljoen, Westerholm & Zheng (2014) study intraday AT on the SPI 200 futures market. Their results show that AT contribute to liquidity and price discovery on this Australian futures market.

Studies also found negative aspects of AT. Grioth (2011) shows that due to the increased intensity of AT, trading based on fundamentals has decreased and subsequently resulted in high levels of stock price volatility. Contrary to Jovanovic & Menvelo (2011) study, Viljoen et al. (2014) argue that HFT's advantage of speed enables them to process information quicker and trade ahead of non-AT investors. This advantage of speed has resulted in adverse selection costs increasing for non-AT investors.

H2: AT intensity decreases during flash crash compared to a normal trading time

Next, we are investigating the impact of AT on liquidity during flash crash with the following hypothesis.

H3: Market liquidity, measured by AT quoted spread, and AT depth, significantly reduces during flash crash compared to a normal trading time

H3.1 AT quoted spread increases during flash crash

H3.2 AT market depth decreases during flash crash

Further, we are looking at the impact of non-AT on market liquidity during a Flash Crash. We propose hypothesis H3 and H4 related to behavioral trading differences (if any) between high frequency algorithmic trading (H3) and non-AT trading (H4). The hypothesis is that market liquidity is also affected negatively by non-AT trades during the flash crash.

H4: Market liquidity, measured by non-AT quoted spread, and non-AT depth, reduces during flash crash compared to a normal trading time

H4.1 Non-AT quoted spread increases during flash crash

H4.2 Non-AT market depth decreases during flash crash

4. DATA AND METHODOLOGY

4.1 Dataset and Sources

This study uses order book and trade data for India's National Stock Exchange (NSE) provided and maintained by DotEx International Limited⁹ that contains a trigger whenever a trade was submitted by an AT. Our dataset is very comprehensive that contains indicator trading mechanism (pre-open or regular), order submission type (new order, Modified order or cancel order), order identification number, time stamp (Jiffies¹⁰), stock symbol (scrip), disclosed volume, type of order (Market, limit, stop), price, flag to identify the order submission channel (Algo, non-algo, Algo through SOR¹¹, and Non-algo through SOR), and client information. We analyze all 500 stocks from the NSE Nifty 500 index from October 4th, 5th and 8th, 2012. These stocks represent the top 500 companies based on market capitalization, totaling 94% of the free

⁹ Dataset related details https://www.nseindia.com/supra_global/content/dotex/data_products.htm.

¹⁰ NSE reports timestamp of an order in Jiffies. The start point of Jiffies count begins from January 01, 1980. One second is equals to 65536 Jiffies.

¹¹ Smart Order Routing (SOR). SOR enable the brokers trading engines to systematically choose the execution destination based on factors viz. price, costs, speed, likelihood of execution and settlement, size, nature or any other consideration relevant to the execution of the order. For details refer to SEBI's circular CIR/MRD/DP/26/2010 issues on August 27, 2010.

float market cap on the NSE. We deleted market orders and limit orders that were “immediate flag” conditioned orders. These orders do not contribute to market liquidity due to immediate execution. To avoid influence of quote stuffing (Egginton et al., 2016), we deleted orders that were canceled within five minutes of order submission and consider orders that are either modified after initial submission or pending in the orderbook. The Flash Crash takes place on October 5th from 9:50hr-10:10hr. Over the three days there are approximately 325 million observations. We also limit ourselves to 500 stocks out of the 1,847 listed on the exchange. Following Aggarwal & Thomas (2013), we cut out the first 15 minutes of trading (pre-open trading session) and stop our analysis at market close (15:30hr). We are left with 642,316 observations.

The dataset used in this paper is unique with detailed information of order-flow and renders quality information to explore orderbook dynamics. Other datasets used in US study did not have the AT indicator; hence proxies are used for studies conducted on U.S markets. Hendershott, Jones and Menkveld (2011) use a normalized measure of NYSE electronic message traffic as a proxy to identify what traders are AT. Hasbrouck & Saar (2013) use a new type of algorithm to proxy for HFT. Scholtus, Dijk and Frijns (2014) use numerous proxies to identify AT including total message activity supplemented by the number of fleeting order proxy.

For studies that do not use proxies to identify AT, there are limitations in the number of stocks in the database. Hendershott and Riordan (2013) dataset has flags down AT orders; however only for 30 stocks. Brogaard, Hendershott and Riordan (2014) use data from 120 randomly selected stocks from a selected amount of HFT firms, with the exclusion of Goldman Sachs and Morgan Stanley.

Our data does not contain restraints for the quantity of companies or the exclusion of HFT firms as aforementioned. Also, the non-existence of market makers and dark pools increases the cleanliness and validity of our data.

4.2 Variable Construction

The variables for AT intensity and market liquidity are calculated following Aggarwal & Thomas (2013).

1. AT-Intensity: Our data is placed into 1-minute buckets, the bucket is used to compact all the data from a given minute. The bucket compares variables from the specific period. AT-intensity is calculated by the total traded volume ($TTV_{ATtrade}$) of AT for a given stock from the 1-minute period divided by the overall total traded volume (AT + Non-AT) (TTV) of that stock and expressed as a percentage. Non-AT intensity is simply 1 minus AT intensity.

$$AT\ Intensity_{i,t} = \frac{TTV_{ATtrade,i,t} \times 100}{TTV_{i,t}} \dots \dots \dots Eq\ (1)$$

2. Quoted spread is calculated by the taking the difference of the best bid and best ask price. It is expressed as a percentage of the mid-quote price (P_{MQ}), which is the average of the best bid ($P_{BestBid}$) and ask price ($P_{BestAsk}$).

$$Quoted\ Spread_{i,t} = \frac{(P_{BestAsk_{i,t}} - P_{BestBid_{i,t}}) \times 100}{P_{MQ_{i,t}}} \dots \dots \dots Eq\ (2)$$

3. Depth is a measure of the total number of share outstanding at either side of the book (Ask side as TSQ and Bid side as TBQ) available for execution at any point of time.

$$Depth_{i,t} = \frac{TSQ_{i,t} + TBQ_{i,t}}{2} \dots \dots \dots Eq\ (3)$$

4.3 Methodology

We use a combination of univariate and multivariate analysis to test the hypotheses. The equations to test the hypotheses 1, 3 and 4 is as below:

$$Liquidity_{i,t} = FC + Control \quad (1)$$

The dependent variable, $Liquidity_{i,t}$, is measured by $Quoted Spread_{i,t}$ or $Depth_{i,t}$ for the overall sample for hypothesis 1. For hypothesis 3 and 4, the variable alternatively is $Quoted Spread_{i,t}$ and $Depth_{i,t}$ for the AT population (hypothesis 3) and non-AT population (hypothesis 4). For all the regressions, independent variable FC is a dummy for flash crash time, and $Control$ is either natural logarithm of market capitalization or natural logarithm of turnover of the stock in the market.

The equation to test hypothesis 2 is as below:

$$AT Intensity_{i,t} = FC + Control \quad (2)$$

The dependent variable, $AT Intensity_{i,t}$, denotes the portion of trades which is algorithmic trading among all orders in the system. Independent variable FC is a dummy for flash crash time, and $Control$ is either natural logarithm of market capitalization or natural logarithm of turnover of the stock in the market.

For robustness, we re-run equation (1) and (2) for five size quintiles of the sample to further understand the behavior of different segments of the market (according to size) and their behavior.

5. RESULTS

Table 1 provides summary statistics for our complete sample. We observe that quoted spread has a mean of 1.71 % for the complete sample and depth is 141,693. Orderbook data indicate that on an average around 86.43% orders are submitted either AT or SOR only less than 14% orders are

from non-algo channels. However, we observe 91.1% of AT orders on the flash crash day. This highlights domination of AT in the NSE. We also note that average AT – Intensity across the study period is 73.47% which indicates that in a Algorithmic traders provide higher volume in the market. Comparison of AT-intensity just before and after the flash crash indicate that AT-intensity Hence, we separate algo trades and non-algo trades to further understand their role in capital markets. When we separate the data by quintiles based on market capitalization we observe that quoted spread decreases from quintile -1 to quintile 5 and depth increases from quintile 1 to quintile -5 which is expected. Similarly we observe that AT – Intensity is highest for quintile – 5, algo trades are highest for large market cap companies.

<Insert Table 1 about Here>

5.1 Flash Crash and Market Liquidity

Our first research question provides a background on the effects suffered by the market during the 20-minute Flash Crash including initial market recovery period. Orders of AT and non-AT are aggregated together to show the overall impact that the Flash Crash had upon the market. Figure 1 presents the mean quoted spread for all stocks in the sample for the flash crash day. We observe a sudden spike between 9:50hr and 10:10hr, as spreads dramatically increased during Flash Crash time.¹² Figure 2 depicts the mean depth of the all the stocks in our sample during October 5th. Reduction in market depth during the flash crash significantly affects market. It took few minutes for depth to recover pre-Flash-Crash.¹³

¹² In appendix Figure 1A we have classified quoted spread by size-quintiles. Each size-quintiles quoted spreads increased dramatically during the Flash Crash time. This graph shows that liquidity deter not only for small sized stocks but also for large market capitalized stocks also. We can observe that smaller stocks spreads have increased disproportionately compared to large stocks

¹³ Figure 2A in appendix shows depth of stocks by size quintiles. A similar pattern is observed for each size quintile where depth has reduced during the flash crash time. Large drop in large cap stocks can be attributed to the volume trading. On the NSE, most of trades are concentrated in fewer large and medium market cap stocks.

<Insert Figure 1 and Figure 2 about here>

5.1.1 Univariate and Multivariate Analysis

Table 2 compares the 20 minutes flash crash on October 5th to the same time (9:50-10:10) on October 4th and 8th using T-Test. Our results show significant changes in our liquidity measures. Quoted spread mean more than doubles on October 5th during flash crash time and market depth decreases by more than 50% during this time in comparison to trading days before and after October 5th. When we look at this results by size quintiles we find that mean quoted spread have increased by more than 3% for small cap stocks (Q1) and by 2.26% for large cap stocks (Q5). These differences are statistically significant at 1 percent level. For depth, we observed that during flash crash time the mean depth of quintile 1 decreased by 1845.7 in comparison to the depth for the same quintile on October 4th and October 8th during the same time. The largest decline in depth is observed in quintile 4, and for each quintile we observe depth has worsen.¹⁴

<Insert Table 2 about here>

To examine the role of algorithmic trading on the market liquidity, we use multivariate regression analysis. Table 3 reports results for full sample and size-quintile based estimation respectively. The dependent variable in Panel A is quoted spread and in Panel B is logarithmic

¹⁴ In table 1A in appendix we show the individual t-tests for liquidity variables for the whole trading day of October 5th and compare it to the mean of liquidity variables on October 4th and October 8th complete trading day. In the first column using all the observation in our sample average quoted spread on Flash Crash Day is higher than the normal trading and depth is lower on flash crash day in comparison to one day before and after this event. The magnitude of difference in table 1A is lower than compared to the magnitude of difference in table 1. In general, we find that average quoted spread has increased by 0.05% and average depth has decreased by 4,516 on October 5th in comparison to October 4th and October 8th, 2012. Our quintile results point out that during the flash crash day large cap stocks (Q5) experienced an increase in 0.20% in quoted spread but small cap stocks (Q1) experienced a decline in their quoted spreads. This results points out that small stocks liquidity is adjusting quickly in contrast to large cap stocks. For depth, small cap stocks (Q1) have shown less decrease in mean depth whereas large cap stocks (Q5) mean depth have decreased greatly. This again shows speed of adjustment of liquidity is better for small stocks in comparison to large stocks.

depth. Our main variable of interest is flash crash dummy. This is a dummy variable which takes the value 1 during the flash crash time between 9:50 am – 10:10 am. Our control variables are Log (Market Cap) to control for size and Log (Turnover) to control rupee volume. Since market cap and turnover are highly correlated, we use this control variables in separate model. We also suppress the intercept. For all observations in Panel A we find that flash crash dummy coefficient is positive and significant when the dependent variable is quoted spread. This indicates that spreads increased during flash crash time which is similar the results obtained in our univariate test and are also visible from the graphs that we have plotted. Our control variables coefficient is negative and significant which is line with literature that quoted spreads share negative relationship with size and turnover. In Panel B, the flash crash dummy has a negative sign and the coefficient is also statistically significant at 1 percent level. This points out that depth decreased during the flash crash time. In table 4, Panel A, flash crash dummy is positive and significant for each size quintiles (Models III- XI). This shows that quoted spread did increase for both small market capitalized and large capitalized stocks. But for depth we find that coefficient of flash crash dummy for small stocks (Q1) have a positive and significant relationship, but for other quintiles the coefficient is negative and significant. This is little different from what we have obtained in our univariate analysis where during flash crash time average depth has decreased significantly for all quintiles. But overall our fixed effects regression for all observations and for quintile observations help us to reject our null hypothesis and we can confidently say that liquidity suffers during flash crash time.

<Insert Table 3 about here>

5.2 Algorithmic Trading: Flash crash and normal trading period

In this section AT behavior will be investigated by testing for AT-intensity. We find that AT intensity significantly changes during the Flash Crash time.

Figure 3 present AT-Intensity during October 5th, the Flash Crash Day. During the Flash Crash time intensity was significantly lower on average than normal.¹⁵ In general, we observe AT – Intensity increases during the trading day and fades away as the trading day comes to an end.

<Insert Figure 3 about here>

5.2.1 Univariate and Multivariate Analysis

Table 4 displays the means and performs univariate test of AT-intensity during the Flash Crash time on October 5th, compared to the same 20 minutes of the two normal trading days before and after the flash crash day. In those 20 minutes, algorithmic trading intensity significantly drops during Flash Crash time from their standard 70-75% level to under 50%. This difference in means for AT intensity is statistically significant at 1 percent level. We observe that AT intensity declines for each quintile which shows that algorithmic traders recede from the market during flash crash time. In smaller stocks (Q1) AT – intensity only decreased by 14 % but for large stock AT – Intensity have decreased by approximately 25 percent.¹⁶

<Insert Table 4 about here>

In table 5 we run multivariate firm fixed effect regression. The dependent variable is AT Intensity. We run the regression on the complete sample and also on different size quintiles. Our

¹⁵ In figure 3A in appendix we present AT – Intensity for October 5th by size quintiles. AT- Intensity decreases for each quintile but the larger decline can be observed for large cap stocks.

¹⁶ In appendix table 2A we compare the mean At-Intensity for October 5th complete trading day with the AT-Intensity of complete trading day on October 4th and October 8th. We find that the decline in AT Intensity is not that substantial. For all observation (column – 1) we find that AT – Intensity only declined by 0. 70%. Across different size quintiles AT – Intensity have not declined by more than 1 percentage. This shows that as soon as the flash crash was over, algorithmic traders were trading normally for the rest of the day.

variable of interest is Flash crash dummy and our control variables are similar to Table 3, Log (Market Cap) and Log (Turnover). In table 5 we find that flash crash dummy coefficient is negative and statistically significant. This result is also similar to the univariate results obtained in table 4 where we observe drastic reduction in AT- Intensity. For each size quintile we find that flash crash dummy coefficient is negative and statistically significant which further verifies our results that during the flash crash algorithmic trading intensity decreases. From the univariate and multivariate results, we observe that during high market stress conditions, AT's abstract from the market. From these results, we can reject the null hypothesis.

<Insert Table 5 about here>

5.3 Algorithmic Trading and Market Liquidity

In this section, AT is tested separately to find the effects they have on market liquidity. The two market liquidity variables were tested on AT's for the 20-minutes during the Flash Crash.

Figure 4 shows the pattern of AT's average quoted spread during the Flash Crash day. On average quoted spread for AT trades increased to 5%-6% during the 20 minutes flash crash time.¹⁷ This pattern resembles the graph in Figure 1 where quoted spread increased during the actual flash crash time. In figure 5 we have plotted AT depth for the whole trading day of October 5th. We observe that average depth for algo trades have declined during the flash crash time.¹⁸ This is again like the graph 3 which represents the mean depth for our complete sample. Overall, we observe from these graphs that algorithmic trades average quoted spread have

¹⁷ In figure 4A in appendix we have plotted AT quoted spread by size quintile. We observe a spike in quoted spread for all size quintiles during those 20 minutes. This pattern resembles the graph in Figure 1A where quoted spread increases during the flash crash time.

¹⁸ In figure 5A we have plotted AT depth by size quintiles. Here also we see that mean depth by each size quintile have decreased. This is comparable to the graph 2A which represents the mean depth by size quintiles.

increased and average depth has decreased during the flash crash time. We also note that market adjustment for liquidity also happens very quickly.

<Insert Figure 5 and Figure 6 about here>

5.3.1 Univariate and Multivariate Analysis

Table 6 compares the differences in means for AT quoted spread and depth during the Flash Crash compared to mean values of October 4th and 8th from 9:50hr-10:10hr. Algorithmic trading average quoted spread is 30.57 percent higher during the Flash Crash time and the mean depth is 55 percent lower. For each size quintile, we observe that quoted spread on October 5th during the flash time is greater than one day before and after the trading day. The largest difference in quoted spread for algo trade is observed for large stocks. Similarly, for depth by each size quintile we observe a decline across different market capitalization.¹⁹

<Insert Table 6 about here>

In table 7, we perform multivariate regression on a sample which contains algorithmic trades only. Our dependent variable is quoted spread in Panel A and log (depth) in Panel B and the main interest variable is Flash Crash Dummy. We control for Market Cap and Turnover in separate models. We find that Flash Crash dummy has a positive relationship with quoted spread and negative relationship to depth for all algo observations (Model I and II). The coefficient for Flash crash dummy for all are statistically significant at 1 percent level. These results verify with our univariate T- test, by suggesting that quoted spread increases during flash crash time and depth has declined for algorithmic traders. We perform similar analysis for quoted spread and

¹⁹ In appendix, table 3A we compare difference in means for AT quoted spread and depth during the entire trading day of October 5th to the entire trading day of October 4th and October 8th. We find that in general for our complete sample quoted spreads have increased by approximately 5 percentage where has depth has fallen by 4 percentage. Across different size quintiles we find that in general quoted spread is higher on October 5th as compared to other days and depth has declined too. Overall, we can observe that the flash crash do carry a major impact across the whole trading day.

depth by size quintiles (Model III – XII). In Panel A across each size quintile we find that the coefficient on Flash Crash dummy is positive and significant suggesting that quoted Spread increased for each size quintile during the Flash Crash time. For Depth in Panel B we find the relationship between depth and flash crash dummy is negative and significant which suggests that depth has declined for algo trades irrespective of size. In general, with our graphs, univariate tests and multivariate tests we find that algorithmic trades suffered liquidity problems during the flash crash time. Their quoted spread increased and depth has decreased. When we look at different size quintiles we observe similar results.

<Insert Table 7 about here>

5.4 Market Liquidity, flash crash and Non-Algorithmic Trading

This section examines the impact of non-algo trading on market liquidity during the flash crash. Figure 6 displays Non-AT quoted spread for all our complete sample on October 5th and Non-AT quoted spread increased to 20 -25 percentage during flash crash time. The pattern for Non-AT quoted spread during Flash Crash day resembles the AT- quoted spread and the overall graph in Figure 1.²⁰ In Figure 7 we find plot the depth of Non-AT trades for the complete market. We observe that during Flash crash time Non-AT trades try to provide depth to the market during flash crash time but also suffer loss of depth during the later stages of the flash crash time. It suggests that Non-AT trade try to support market depth during flash crash time and as AT- trades started adjusting to the new environment they started retreating from the market. The pattern for Non-AT depth in figure 13 is different from what we have observed so far. One thing that

²⁰ Figure 6A in appendix we plot quoted spread by size quintiles for Non- Algo observations. We observe that Non-At quoted spread have increased for each quintile irrespective of the size. This behaviour for Non- At quoted spread during flash crash day resembles AT- quoted spread by size quintiles in graph 4A and the complete sample observation by size quintiles in graph 1A

remains constant is that after Non -AT depth reduces after market recovery and this also occurs very quickly. This is like AT trades which also recover quickly after the flash crash.²¹

<Insert Figure 6 and 7 about here>

5.4.1 Univariate and Multivariate Analysis

We perform T-Test and firm fixed effects next to get a better understanding of Non-AT Trades during flash crash time. Table 8 presents the T-Tests where we compare the means of quoted spread and depth for Non -Algo trades during the 20 minutes of flash crash time on October 5th, 2012. For each variable, we have compared the means for the all Non- Algorithmic trades together and by size quintiles. Here we find that quoted spreads have increased by 100% during those 20 minutes for all Non –Algorithmic trade observations. This is true for each size quintiles also. This points out that Non- Algorithmic trades quoted spreads widened greatly irrespective of market capitalization. For depth, we find that Non-Algorithmic trades depth increased during the Flash crash time. This points out that Non-Algorithmic traders try to provide additional depth during crisis period. Looking at the results by size quintiles, small cap stock (Q1, Q2, Q3) depth increased whereas large cap stock (Q4, Q5) depth decreased during these 20 minutes. Large cap stocks behavior for Non-Algorithmic trades resembles Algorithmic Trades, that they retreated depth from the market, but small cap stock non-algorithmic trades behave differently is notable finding.²²

²¹ Figure 7A in appendix we plot depth by size quintiles for Non- Algo observations. We observe similar pattern as observed in Figure 7.

²² In appendix table 4A we perform the t-test on quotes and depth for non-algorithmic trades on complete trading day of October 5th and compare them to non-algorithmic quotes and depth on October 4th and October 8th. Overall, we find that quoted spread did increase by 5 percent for non-algorithmic trades on October 5th. We also note that for non -algorithmic trade, increase in quoted spread in not universal across different size quintiles. For small cap stocks (Q1) we find the difference in means is negative and statistically significant but for large cap stocks the difference in

<Insert Table 8 about here>

We have also performed multivariate analysis similarly to those performed for algorithmic traders. Our sample consists of non-algo trades only. The main dependent variables are quoted spread and log (depth) and the independent variable of interest is Flash Crash Dummy. We also control for size and turnover by taking their logarithmic values. In Panel A of table 9 we run the firm fixed effects model for quoted spread and in Panel B we run firm fixed effects for log (Depth). We run firm fixed effects model by size quintile for both liquidity variables.

For the complete sample of non algo trades (Model I and II, in Panel A & B) we find the coefficient of flash crash dummy positive and significant for both quoted spread and depth. These results confirm that quoted spread for all types of trades have increased and deteriorated during flash crash time. But for depth, we can now confirm that non-algo trades try to support the market in stress situation For quintile results (Model III-XII, in Panel A & B) we find that flash crash dummy is positive across all models for quoted spread and depth which further adds that regardless of the size non-algorithmic trades quoted spread increased during the flash crash time and they provided depth during the flash crash time.

<Insert Table 9 about here>

means is positive and statistically significant. Also for depth we find that for all observation Non-algorithmic traders provided higher depth on October 5th than other trading days. When we separate our sample by size quintile, different quintiles have varied results. For quintile Q1, Q3 and Q5 we see that depth is higher on October 5th but for other quintiles depth is lower on October 5th than other trading days. Overall, we can say that Non- Algorithmic traders quoted spread were higher on October 5th which is similar to Algorithmic trades and overall market. Non- Algorithmic provided depth in the market when Algorithmic traders started retreating depth from the market.

Overall, we see that quoted spread has declined for the whole market notwithstanding the type of trade. Algorithmic traders reduced their intensity and started withdrawing their trading orders during flash crash time thus reducing depth in the market. Non-algorithmic traders increased their trading orders by trying to support the market depth. Our results are robust across univariate t-test and multivariate firm fixed effects regression.

6. CONCLUSION

We examine the impact of algorithmic trading on market liquidity, market quality during the flash crash on October 5, 2012 in the NSE. This study use unique orderbook data that allows to explicitly examine the impact of orders submitted through algo and non-algo channels on the exaggerating the panic on the flash crash day. We also examine the algo trading dynamics during the normal trading periods and compare with the flash crash event.

The findings show that quoted spread increases and market depth decreases during flash crash time, compared to other trading time. Which means, the market liquidity decreases during flash crash time. Also flash crash time observes a significant reduction in AT-intensity. We also find that AT affects market liquidity negatively during flash crash and non-AT provides more liquidity to the market during flash crash time, compared to AT. The results are robust when we divide the sample into 5 quintiles based on market capitalization of the stock (size quintiles). From the analysis we conclude that the Flash Crash did not cause damage to the market that sustained beyond the 20 minute period.

We make several contributions to the existing literature. This paper was the first to look at a Flash Crash event outside of the May 2010 U.S crash. Adding another perspective adds to the depth of research of Flash Crashes. Furthermore, we have breached the norm by researching an

emerging market. Datasets that contain a trigger for AT's are rare. The unique dataset used in this study that identifies the trader has provided a clear picture of how AT and non-AT acted during the Flash Crash. The new insight contributed by this paper enables a deeper understanding of market structure and behavior characteristic of AT and Non-AT. Finally, with the utilization of our fixed effect models and breaking the sample into quintiles we extend the methodology used in the literature on Flash Crashes.

This study has implications upon numerous groups who participate and regulate within the market. Investors in India and outside emerging markets will find this research insightful to understand AT behavior within the market. Further awareness of AT will assist investors in their investment decisions by decreasing adverse selection costs.

As events, similar to the Flash Crash become more prevalent in different types of markets across the financial world, regulators need to be aware of the consequences and danger of the event when algorithmic high frequency trading is allowed. The field of Flash Crash literature is currently developing and conclusive findings can have significant implications on how AT's are regulated in the future. The benefits of algorithms in the market place are compelling however, ensuring that regulators prevent unnecessary market volatility caused by AT's needs to be a priority as the financial systems continues to advance.

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Tables and Figures

Table 1: Summary Statistics

In this table, we provide the summary statistics of variables that are used in this article. Quoted spread is calculated by the taking the difference of the best bid and best ask price. It is expressed as a percentage of the mid-quote price), which is the average of the best bid and ask price. Depth is a measure of the total number of share outstanding at either side of the book available for execution at any point of time. AT Intensity is calculated by the total traded volume of AT for a given stock divided by the overall total traded volume of that stock. Quintile 1 is the summary statistics for smallest market cap stock and Quintile 5 summary statistics for largest market cap stocks. Std Dev is the standard deviation of each variable.

Observations	Variable	N	Mean	Std Dev	Minimum	Maximum
All Observations	Quoted Spread (%)	642316	1.7108	2.5116	0.00	39.83
	Depth	642316	141692.6800	524136.2300	1.00	19942831.50
	Market Capitalization (Rs Mn)	638004	183414.9800	389214.7000	328.21	2775555.77
	Turnover (Rs Mn)	638004	441.1258	1700.5400	0.01	45389.78
	AT Intensity (%)	290016	73.4773	28.2566	0.00	100.00
Quintile 1	Quoted Spread (%)	127728	2.2431	3.1845	0.00	39.83
	Depth	127728	9374.8900	45705.7000	1.00	4332815.00
	Market Capitalization (Rs Mn)	127728	9412.7100	4177.5200	328.21	16003.30
	Turnover (Rs Mn)	127728	52.8600	79.0535	0.01	612.08
	AT Intensity (%)	49816	52.4512	30.2828	0.00	99.98
Quintile 2	Quoted Spread (%)	127531	1.8880	2.5876	0.00	31.50
	Depth	127531	44537.5900	203479.1700	1.00	12314236.50
	Market Capitalization (Rs Mn)	127531	23846.0600	4577.7300	16167.60	32644.67
	Turnover (Rs Mn)	127531	131.9041	209.8854	0.25	1349.25
	AT Intensity (%)	56381	65.5294	29.0988	0.01	99.98
Quintile 3	Quoted Spread (%)	127279	1.7557	2.5670	0.00	39.25
	Depth	127279	59815.2500	225526.6800	1.00	7509914.50
	Market Capitalization (Rs Mn)	127279	49025.3300	12408.5300	32682.80	76790.71
	Turnover (Rs Mn)	127279	171.5772	425.6446	0.44	3920.54
	AT Intensity (%)	57046	70.6398	27.9640	0.03	100.00
Quintile 4	Quoted Spread (%)	127606	1.5029	2.2329	0.00	32.93
	Depth	127606	180692.6600	599700.5400	1.00	11926300.50
	Market Capitalization (Rs Mn)	127606	126699.9400	38799.5600	78305.55	231634.23
	Turnover (Rs Mn)	127606	430.9534	759.6937	1.75	6303.89
	AT Intensity (%)	60945	81.3687	22.4503	0.01	100.00
Quintile 5	Quoted Spread (%)	127860	1.1821	1.6315	0.00	30.07
	Depth	127860	409225.4000	902826.6500	1.00	19942831.50
	Market Capitalization (Rs Mn)	127860	706777.2700	635195.8800	235623.88	2775555.77
	Turnover (Rs Mn)	127860	1415.8900	3514.9600	14.65	45389.78
	AT Intensity (%)	63662	91.2719	13.6495	0.16	100.00

Table 2: T – Test during Flash Crash Time

In this table, we compare the means of quoted spread and depth during the actual flash crash on October 5th between 9:50 hr to 10:10h with the means of these variables on October 4th and October 8, one day before and after the flash crash at the same corresponding time. T- Test is implemented on the complete dataset (ALL OBS) and also on size quintiles. ***, ** is the statistical significance at 1% and 5% respectively

Quoted Spread (%)						
Date and Time	All OBS	Q1	Q2	Q3	Q4	Q5
OCT 5th 9:50 am -10:10 am	4.6860	6.1555	4.9096	4.6494	3.8963	3.7272
OCT 4 and 8- 9:50 am - 10:10 am	2.2295	3.1403	2.3723	2.2383	1.8103	1.4674
Difference in Means	2.4565***	3.0152***	2.5373***	2.4111***	2.0860***	2.2598***
Standard Error	(0.056)	(0.154)	(0.126)	(0.122)	(0.109)	(0.092)
Depth						
Date and Time	All OBS	Q1	Q2	Q3	Q4	Q5
OCT 5th 9:50 am -10:10 am	64518.4	8894.9	27347.8	36611.5	72523.8	185577
OCT 4 and 8- 9:50 am - 10:10 am	141035	10740.6	45123.8	63456.4	210419	404632
Difference in Means	-76516.4***	-1845.7**	-17776***	-26844.9***	-137895***	-219055***
Standard Error	(6710.3)	(1446.4)	(4779.6)	(6489)	(21096.7)	(21096.7)

Table 3: Regression Results for All Observations and Size Quintile

In this table, we performed firm fixed effects on all observations and size quintiles. Quoted Spread is the dependent variable in Panel A. Log (Depth) is the dependent variable in Panel B. Flash Crash Dummy is a dummy variable taking the value 1 when the actual flash crash happened on October 5th, 2012 between 9:50 am – 10:10 am and 0 otherwise. Log (Market Cap) controls for the size of the firm and Log (Turnover) control for rupee turnover. ***, **, * represents statistical significance at 1%, 5 % and 10% respectively.

Panel A: Quoted Spread												
Parameters	Quoted Spread (%)											
	I- All OBS	II-All OBS	III-Q1	IV-Q1	V-Q2	VI-Q2	VII-Q3	VIII-Q3	IX-Q4	X-Q4	XI-Q5	XII-Q5
Flash Crash Dummy	2.9435*** (0.032)	2.9505*** (0.032)	3.7827*** (0.089)	3.7758*** (0.089)	2.9765*** (0.076)	2.9742*** (0.076)	2.9112*** (0.074)	2.9198*** (0.075)	2.3878*** (0.066)	2.3783*** (0.066)	2.5729*** (0.048)	2.5717*** (0.048)
Log (Market Cap)	-3.1862*** (0.198)		-4.3178*** (0.496)		-2.1081*** (0.417)		-2.9786*** (0.559)		-3.7224*** (0.428)		-2.6179*** (0.342)	
Log (Turnover)		-0.2040** (0.006)		-0.3805*** (0.016)		-0.1844*** (0.015)		-0.1671** (0.015)		-0.2009*** (0.015)		-0.0161* (0.009)
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No of Observation	638004	638004	127728	127728	127531	127531	127279	127279	127606	127606	127860	638004
Adjusted R Square	0.1084	0.1095	0.1071	0.1105	0.0834	0.0843	0.0763	0.0771	0.0836	0.0844	0.0926	0.1095

Panel B Log (Depth)												
Parameters	Log (Depth)											
	I- All OBS	II-All OBS	III-Q1	IV-Q1	V-Q2	VI-Q2	VII-Q3	VIII-Q3	IX-Q4	X-Q4	XI-Q5	XII-Q5
Flash Crash Dummy	-0.3217*** (0.023)	-0.3368*** (0.023)	0.4357*** (0.045)	0.4248*** (0.045)	-0.1690*** (0.046)	-0.1674*** (0.046)	-0.2817*** (0.047)	-0.3036*** (0.047)	-0.8060*** (0.053)	-0.7878*** (0.052)	-0.8522*** (0.062)	-0.9335*** (0.062)
Log (Market Cap)	6.7200*** (0.139)		12.1922*** (0.252)		6.0497** (0.253)		5.9917*** (0.353)		4.6693*** (0.339)		4.6021*** (0.437)	
Log (Turnover)		0.4489*** (0.005)		0.5833*** (0.008)		0.3819*** (0.009)		0.4264** (0.010)		0.3436*** (0.011)		0.4428*** (0.012)
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No of Observation	638004	638004	127728	127728	127531	127531	127279	127279	127606	127606	127860	638004
Adjusted R Square	0.6700	0.6738	0.5393	0.5495	0.6595	0.6624	0.6741	0.6783	0.6215	0.6234	0.3801	0.3857

Table 4: T – Test during Flash Crash Time

In this table, we compare the mean of AT-Intensity during the actual flash crash on October 5th between 9:50 hr to 10:10 hr with the mean of this variable on October 4th and October 8, one day before and after the flash crash at the same corresponding time. T- Test is implemented on the complete dataset (ALL OBS) and on size quintiles. ***, ** is the statistical significance at 1% and 5% respectively

AT- Intensity (%)						
Date and Time	All OBS	Q1	Q2	Q3	Q4	Q5
OCT 5th 9:50 am -10:10 am	48.3488	34.8132	38.0719	39.8923	55.1807	65.1859
OCT 4 and 8- 9:50 am - 10:10 am	70.7243	48.9211	62.0890	68.7592	80.1284	90.8771
Difference in Means	-22.3755***	-14.1079***	-24.0171***	-28.8669***	-24.9477***	-25.6913***
Standard Error	(0.722)	(1.753)	(1.668)	(1.5805)	(1.311)	(0.9083)

Table 5: Regression Results for All Algo -trades observations and size quintile

In this table, we performed firm fixed effects on all observations and every size quintiles. At Intensity is the dependent variable. Flash Crash Dummy is a dummy variable taking the value 1 when the actual flash crash happened on October 5th, 2012 between 9:50 am – 10:10 am and 0 otherwise. Log (Market Cap) controls for the size of the firm and Log (Turnover) control for rupee turnover. ***, **, * represents statistical significance at 1%, 5 % and 10% respectively.

Parameters	AT- Intensity											
	I- All OBS	II-All OBS	III-Q1	IV-Q1	V-Q2	VI-Q2	VII-Q3	VIII-Q3	IX-Q4	X-Q4	XI-Q5	XII-Q5
Flash Crash Dummy	-27.3717*** (0.421)	-27.3571*** (0.421)	-16.7588*** (1.361)	-16.9257*** (1.359)	-29.0993*** (1.155)	-29.1255*** (1.154)	-32.2458*** (0.992)	-32.2475*** (0.993)	-29.2929*** (0.791)	-29.2728*** (0.791)	-27.1751*** (0.432)	-26.9512*** (0.432)
Log (Market Cap)	-16.2256*** (2.320)		63.3707*** (6.278)		-34.9001*** (5.5254)		-52.4092*** (6.626)		-50.4094*** (4.626)		-13.8332** (2.961)	
Log (Turnover)		-0.0358*** (0.079)		2.9421*** (0.212)		-2.4300* (0.218)		0.5877*** (0.189)		-0.6442*** (0.168)		-1.1470*** (0.084)
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No of Observation	287850	287850	49816	49816	56381	56381	57046	57046	60945	60945	63662	63662
Adjusted R Square	0.5616	0.5616	0.3570	0.3581	0.4225	0.4234	0.5021	0.5017	0.4790	0.4781	0.5171	0.5184

Table 6: T – Test during Flash Crash Time for Algo Trades

In this table, we compare the means of quoted spread and depth of algo trades during the actual flash crash on October 5th between 9:50 hr to 10:10 hr with the means of these variables on October 4th and October 8, one day before and after the flash crash at the same corresponding time. T- Test is implemented on the complete dataset (ALL OBS) and on size quintiles. ***, **, * is the statistical significance at 1%, 5% and 10% respectively

Quoted Spread (%)- Algo Trades						
Date and Time	All OBS	Q1	Q2	Q3	Q4	Q5
OCT 5th 9:50 am -10:10 am	1.3779	1.4774	1.2509	1.1910	0.9580	1.9550
OCT 4 and 8- 9:50 am - 10:10 am	1.0722	1.3780	1.1586	1.1398	0.9415	0.7419
Difference in Means	0.3057***	0.0994*	0.0924**	0.0513	0.0165	1.2131***
Standard Error	(0.024)	(0.055)	(0.0422)	(0.044)	(0.037)	(0.073)

Depth- Algo Trades						
Date and Time	All OBS	Q1	Q2	Q3	Q4	Q5
OCT 5th 9:50 am -10:10 am	119174	6979.2	30423.5	46778.5	125968	324028
OCT 4 and 8- 9:50 am - 10:10 am	269721	15866.8	77061.8	110727	391864	754769
Difference in Means	-150098***	-8887.6***	-46638.3***	-63948.9***	-265896***	-430741***
Standard Error	(14619.5)	(3509.2)	(10628.9)	(14225.4)	(42961.2)	(44082)

Table 7: Regression Results for Algo-Trades for all observations and by size quintile

In this table, we performed firm fixed effects on every size quintiles for algo-trades. Quoted Spread is the dependent variable in Panel A. Log (Depth) is the dependent variable in Panel B. Flash Crash Dummy is a dummy variable taking the value 1 when the actual flash crash happened on October 5th, 2012 between 9:50 am – 10:10 am and 0 otherwise. Log (Market Cap) controls for the size of the firm and Log (Turnover) control for rupee turnover. ***, **, * represents statistical significance at 1%, 5 % and 10% respectively.

Panel A: Quoted Spread												
Parameters	Algo Trades Observation- Quoted Spread (%)											
	I- All OBS	II-All OBS	III-Q1	IV-Q1	V-Q2	VI-Q2	VII-Q3	VIII-Q3	IX-Q4	X-Q4	XI-Q5	XII-Q5
Flash Crash Dummy	0.3473*** (0.013)	-1.6862*** (0.026)	0.1674*** (0.038)	0.1684*** (0.038)	0.0857*** (0.033)	0.0874*** (0.033)	0.0920*** (0.028)	0.0934*** (0.028)	0.0094 (0.023)	0.0093*** (0.023)	1.2045*** (0.021)	1.1954*** (0.021)
Log (Market Cap)	-1.8337*** (0.071)		-0.8484*** (0.177)		-1.4690*** (0.161)		-1.0679*** (0.186)		-5.0021*** (0.140)		-1.5486*** (0.145)	
Log (Turnover)		-0.4115*** (0.002)		-0.0789*** (0.006)		-0.0499*** (0.006)		-0.0412** (0.005)		-0.0964*** (0.005)		-0.0275*** (0.004)
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No of Observation	316141	316141	59614	59614	63198	63198	63856	63856	64836	64836	64637	64637
Adjusted R Square	0.3436	0.3431	0.3496	0.3514	0.2462	0.2461	0.3046	0.3050	0.2994	0.2900	0.3081	0.3074

Panel B: Log (Depth)												
Parameters	All Observation- Log (Depth)											
	I- All OBS	II-All OBS	III-Q1	IV-Q1	V-Q2	VI-Q2	VII-Q3	VIII-Q3	IX-Q4	X-Q4	XI-Q5	XII-Q5
Flash Crash Dummy	-1.6658*** (0.027)	-1.6862*** (0.027)	-0.7543*** (0.074)	-0.7946*** (0.073)	-1.4708*** (0.066)	-1.4749*** (0.065)	-1.7080*** (0.061)	-1.7202*** (0.060)	-2.0441*** (0.053)	-2.0324*** (0.052)	-2.0771*** (0.044)	-2.1434*** (0.043)
Log (Market Cap)	5.3027*** (0.148)		14.2952*** (0.349)		4.8099*** (0.320)		2.5941*** (0.406)		1.6366*** (0.319)		2.0160*** (0.308)	
Log (Turnover)		0.4414*** (0.005)		0.6989*** (0.011)		0.2676*** (0.011)		0.3882*** (0.011)		0.2830*** (0.011)		0.3312*** (0.009)
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No of Observation	316141	316141	59614	59614	63198	63198	63856	63856	64836	64836	64637	64637
Adjusted R Square	0.8560	0.8587	0.6847	0.6961	0.7755	0.7764	0.8205	0.8241	0.8483	0.8498	0.7772	0.7821

Table 8: T – Test during Flash Crash Time for Non-Algo Trades

In this table, we compare the means of quoted spread and depth of non-algo trades during the actual flash crash on October 5th between 9:50 hr to 10:10 hr with the means of these variables on October 4th and October 8, one day before and after the flash crash at the same corresponding time. T- Test is implemented on the complete dataset (ALL OBS) and on size quintiles. ***, **, * is the statistical significance at 1%, 5% and 10% respectively.

Quoted Spread (%) - Non-Algo Trades						
Date and Time	All OBS	Q1	Q2	Q3	Q4	Q5
OCT 5th 9:50 am -10:10 am	6.9604	8.4462	7.1590	6.8226	6.2770	5.4994
OCT 4 and 8- 9:50 am - 10:10 am	3.2854	4.5117	3.4856	3.2389	2.6642	2.1987
Difference in Means	3.6749***	3.9345***	3.6734***	3.5837***	3.6128***	3.3007***
Standard Error	(0.091)	(0.227)	(0.198)	(0.195)	(0.185)	(0.156)

Depth- Non-Algo Trades						
Date and Time	All OBS	Q1	Q2	Q3	Q4	Q5
OCT 5th 9:50 am -10:10 am	26943	9833	25456.8	30222.5	29220.4	47125.5
OCT 4 and 8- 9:50 am - 10:10 am	24033.7	6751.3	15826.3	20403.9	32089.3	51680.9
Difference in Means	2909.3**	3081.7***	9630.6***	9818.6***	-2868.9	-4555.4
Standard Error	(1461.6)	(896.1)	(2091.7)	(2973.2)	(4119.4)	(5641.5)

Table 9: Regression Results for Non- Algo-Trades for all observations and by size quintile

In this table, we performed firm fixed effects on every size quintiles for non-algo-trades. Quoted Spread is the dependent variable in Panel A. Log (Depth) is the dependent variable in Panel B. Flash Crash Dummy is a dummy variable taking the value 1 when the actual flash crash happened on October 5th, 2012 between 9:50 hr – 10:10 hr and 0 otherwise. Log (Market Cap) controls for the size of the firm and Log (Turnover) control for rupee turnover. ***, **, * represents statistical significance at 1%, 5 % and 10% respectively.

Panel A: Quoted Spread												
Parameters	Non-Algo Trades Observation- Quoted Spread (%)											
	I- All OBS	II-All OBS	III-Q1	IV-Q1	V-Q2	VI-Q2	VII-Q3	VIII-Q3	IX-Q4	X-Q4	XI-Q5	XII-Q5
Flash Crash Dummy	4.4186*** (0.055)	4.4248*** (0.055)	5.1183*** (0.138)	5.0973*** (0.138)	4.429*** (0.125)	4.4196*** (0.125)	4.2618*** (0.125)	4.2796*** (0.125)	4.1025 (0.117)	4.0851*** (0.117)	3.9169*** (0.090)	3.9227*** (0.090)
Log (Market Cap)	-4.2219*** (0.361)		-6.8248*** (0.860)		-2.7206*** (0.759)		-4.2750*** (1.019)		-2.7112*** (0.800)		-3.3847*** (0.638)	
Log (Turnover)		-0.3345*** (0.012)		-0.5959*** (0.028)		-0.3089*** (0.028)		-0.2742** (0.029)		-0.2926*** (0.028)		-0.0544*** (0.018)
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No of Observation	321863	321863	68114	68114	64333	64333	63423	63423	62770	62770	63223	63223
Adjusted R Square	0.1483	0.1501	0.1393	0.1443	0.1164	0.1179	0.1272	0.1272	0.1376	0.1389	0.1169	0.1167

Panel B: Log (Depth)												
Parameters	Non Algo Trade Observation- Log (Depth)											
	I- All OBS	II-All OBS	III-Q1	IV-Q1	V-Q2	VI-Q2	VII-Q3	VIII-Q3	IX-Q4	X-Q4	XI-Q5	XII-Q5
Flash Crash Dummy	0.6175*** (0.023)	0.6059*** (0.023)	0.9189*** (0.049)	0.9198*** (0.048)	0.6859*** (0.050)	0.6947*** (0.049)	0.7233*** (0.050)	0.6923*** (0.050)	0.2718*** (0.052)	0.2973*** (0.052)	0.3372*** (0.056)	0.2399*** (0.055)
Log (Market Cap)	8.2541*** (0.151)		10.4808*** (0.305)		7.2618*** (0.302)		9.5907*** (0.411)		8.2115*** (0.359)		7.4090*** (0.398)	
Log (Turnover)		0.4977*** (0.005)		0.5013*** (0.010)		0.4941*** (0.011)		0.4785*** (0.012)		0.4290*** (0.013)		0.5647*** (0.011)
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No of Observation	321863	321863	68114	68114	64333	64333	63423	63423	62770	62770	63223	63223
Adjusted R Square	0.6788	0.6858	0.5132	0.5233	0.6812	0.6879	0.7014	0.7069	0.6645	0.6678	0.5580	0.5732

Figures

Figure 1: Quoted Spread on October 5th, 2012

The horizontal axis represent time (in hrs) and the vertical axis represents quoted spread in percentage.

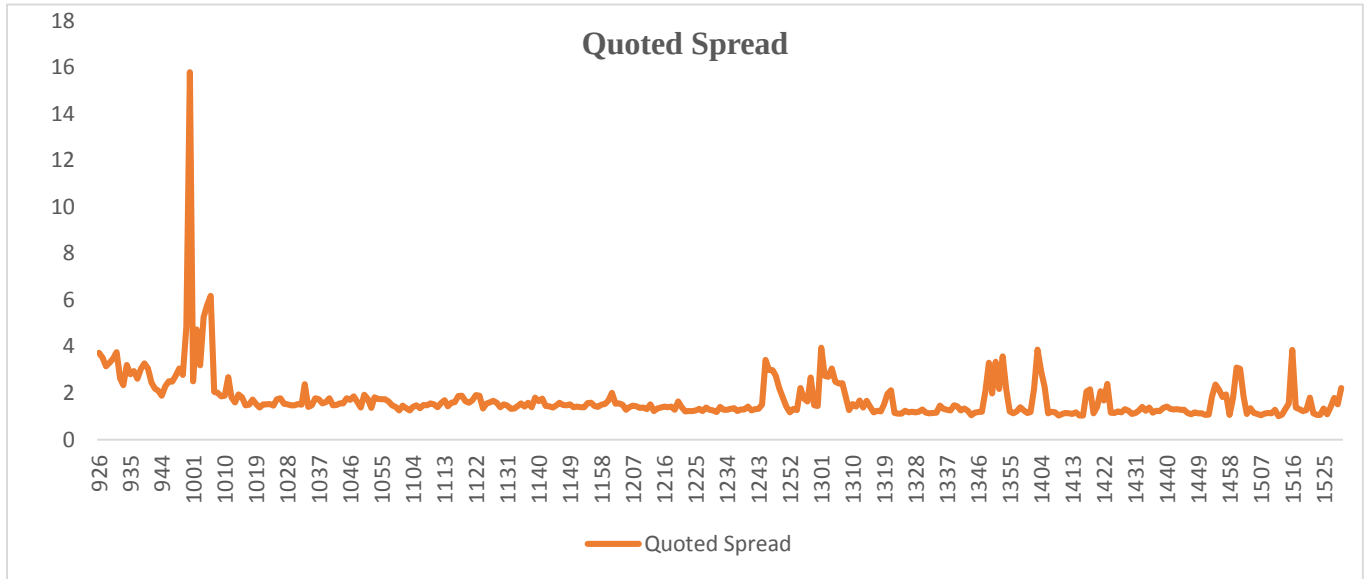


Figure 2: Depth on October 5th, 2012

The horizontal axis represent time (in hrs) and the vertical axis represents Depth.

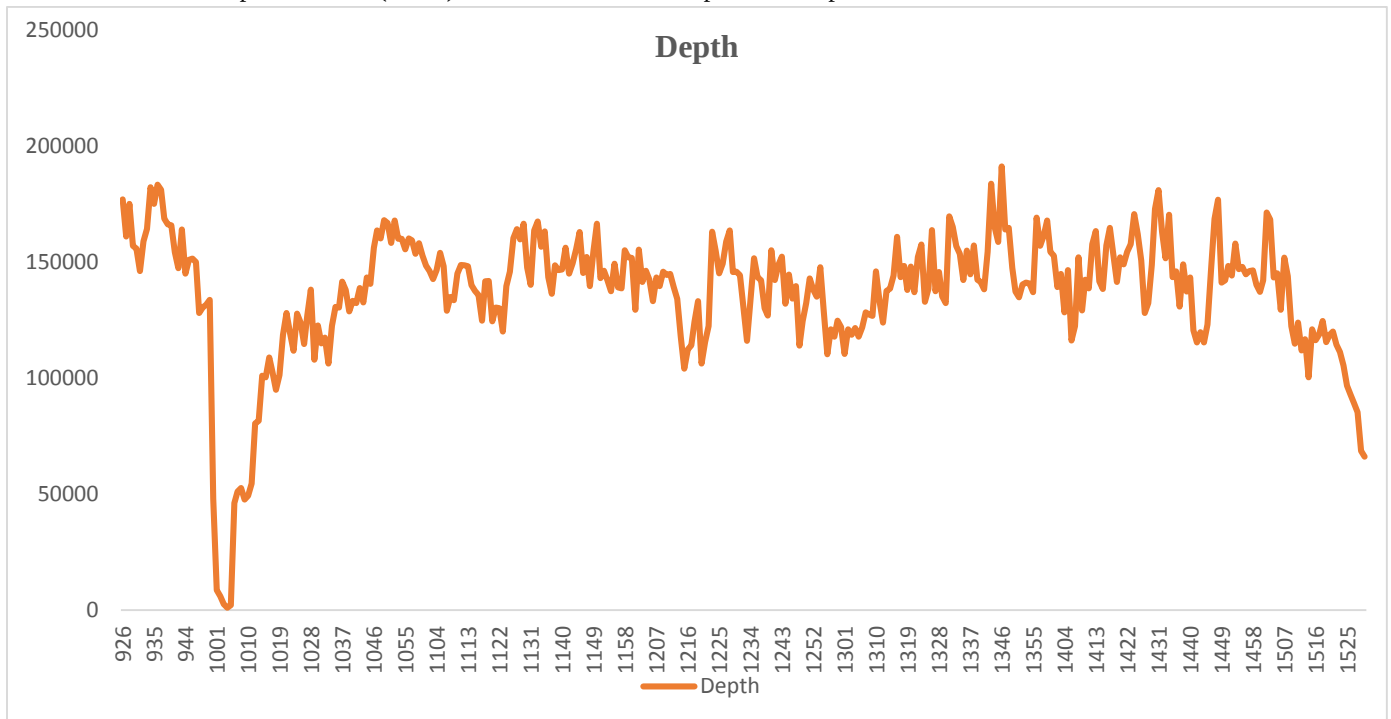


Figure 3: AT-Intensity (%) on October 5th, 2012

The horizontal axis represent time (in hrs) and the vertical axis represents At-Intensity in percentage.

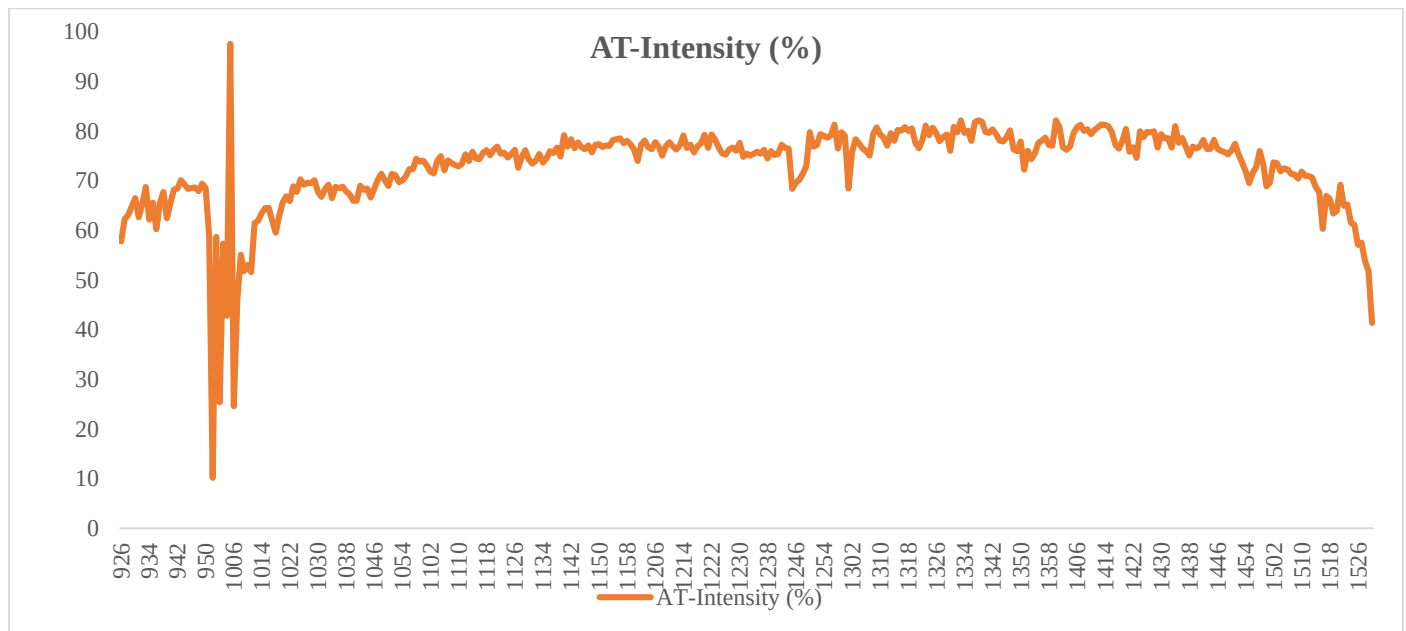


Figure 4: Quoted Spread on October 5th, 2012 for Algo-Trades

The horizontal axis represent time (in hrs) and the vertical axis represents quoted spread in percentage.

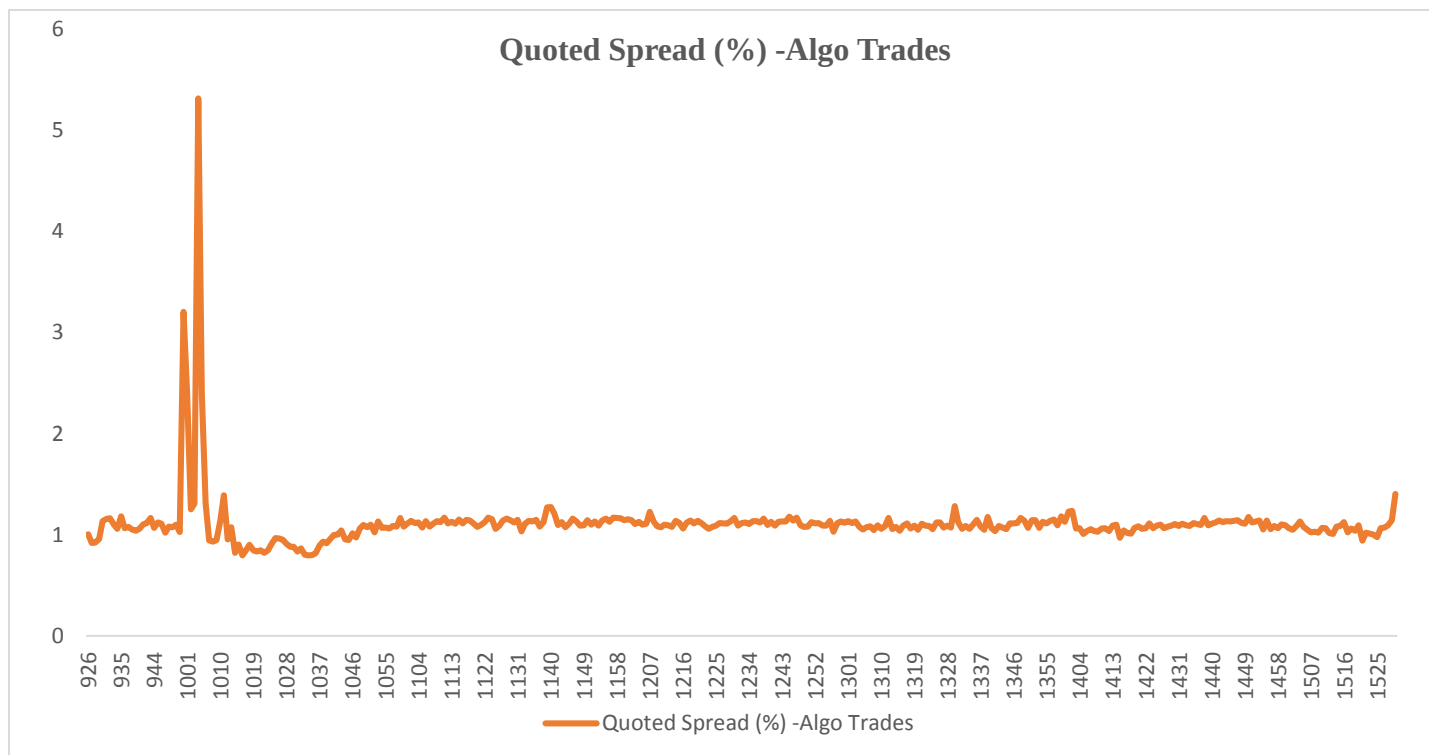


Figure 5: Depth on October 5th, 2012 for Algo Trades

The horizontal axis represent time (in hrs) and the vertical axis represents Depth.

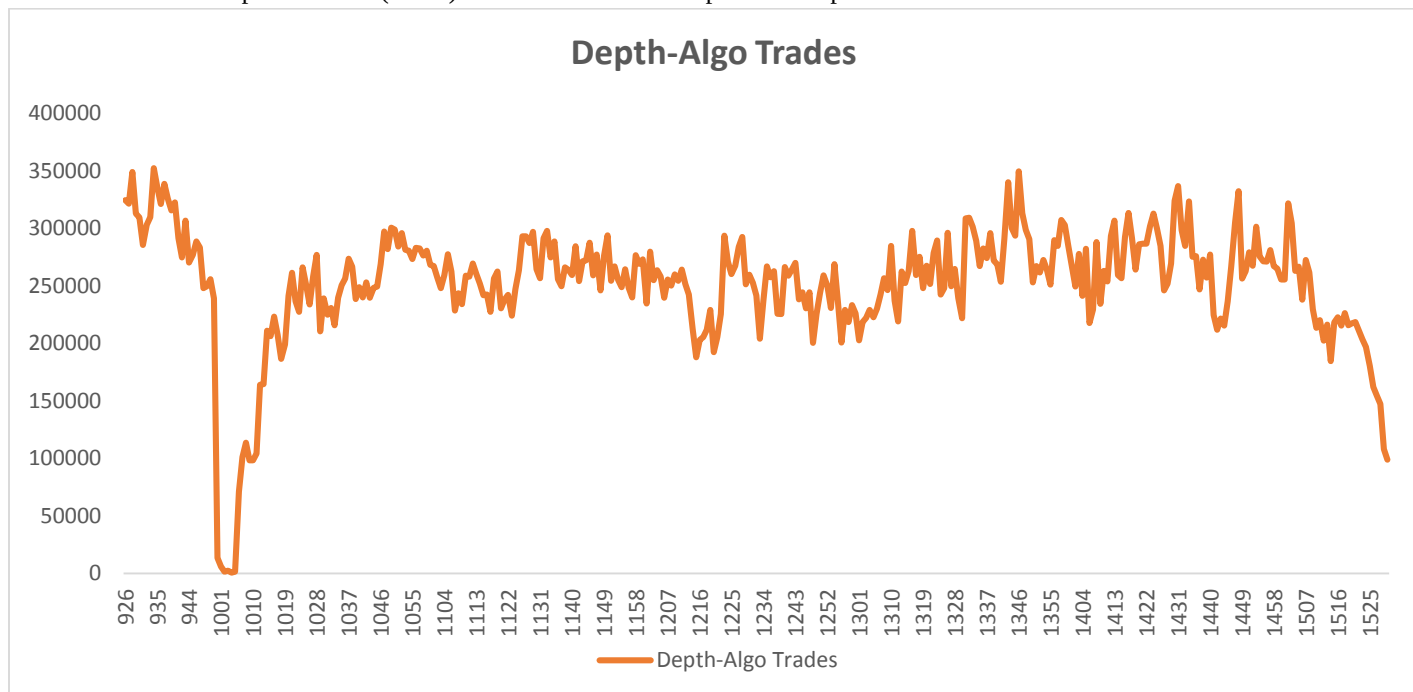


Figure 6: Quoted Spread on October 5th, 2012 for Non Algo-Trades

The horizontal axis represent time (in hrs) and the vertical axis represents quoted spread in percentage.

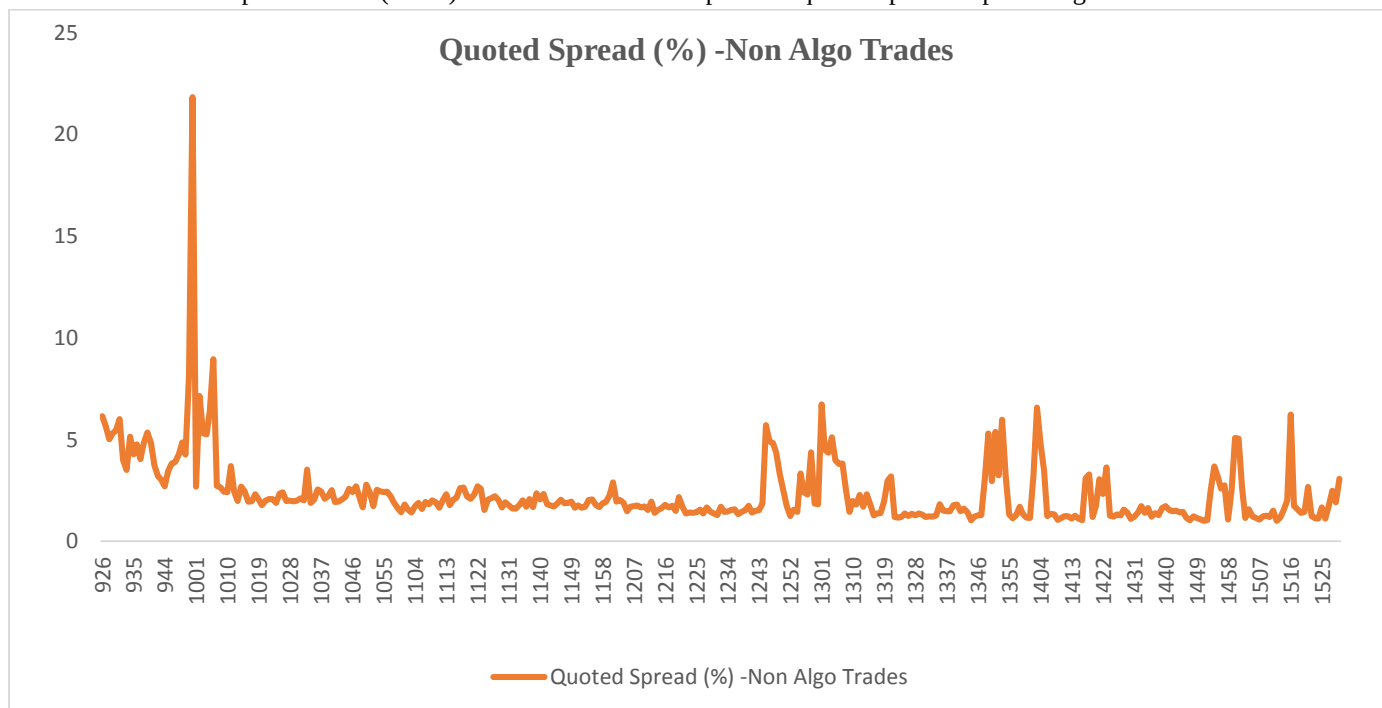
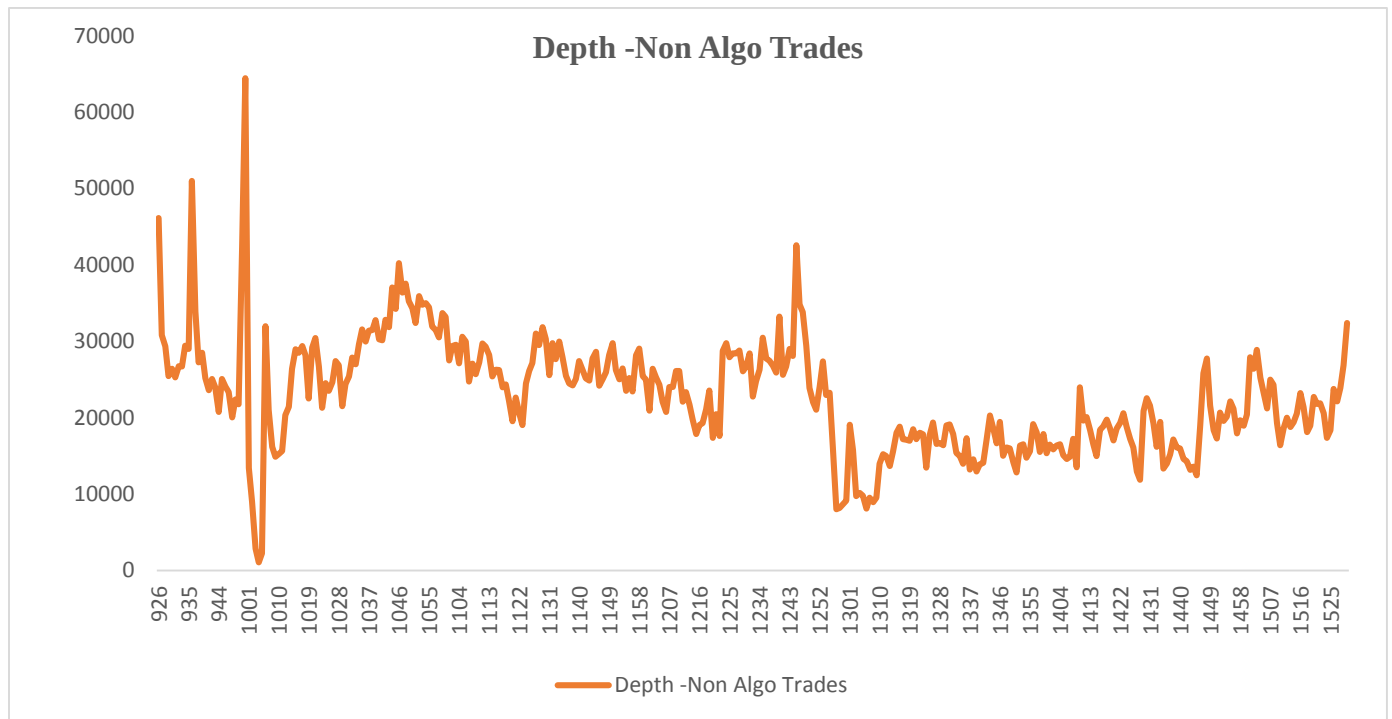


Figure 7: Depth on October 5th, 2012 for Non-Algo Trades

The horizontal axis represent time (in hrs) and the vertical axis represents Depth.



Appendix

Table 1A: T – Test during Flash Crash Day

In this table, we compare the means of quoted spread and depth during the whole trading day of October 5th (Flash Crash Day) with the means of these variables on October 4th and October 8, one day before and after the flash crash for the entire trading day. T- Test is implemented on the complete dataset. (ALL OBS) and on size quintiles. ***, ** is the statistical significance at 1% and 5% respectively

Quoted Spread (%)						
Date and Time	All OBS	Q1	Q2	Q3	Q4	Q5
OCT 5th	1.7451	2.1856	1.9220	1.7496	1.5659	1.3206
OCT 4 and OCT 8	1.6940	2.2713	1.8717	1.7588	1.4729	1.1141
Difference in Means	0.0511***	-0.0857***	0.0503***	-0.0092	0.0931***	0.2065***
Standard Error	(0.007)	(0.019)	(0.016)	(0.152)	(0.013)	(0.010)

Depth						
Date and Time	All OBS	Q1	Q2	Q3	Q4	Q5
OCT 5th	138659	8900.5	43945.6	57445.7	152493	426800
OCT 4 and OCT 8	143175	9607.1	44822.1	61012.9	194121	400596
Difference in Means	-4515.9***	-706.6***	-876.5	-3567.3***	-41627.4***	-26203.2***
Standard Error	(1392.8)	(272.2)	(1216.9)	(1338.6)	(3589.4)	(5372)

Table 2A: T – Test during Flash Crash Day

In this table, we compare the mean of AT-Intensity during the whole trading day of October 5th (Flash Crash Day) with the mean of this variable on October 4th and October 8, one day before and after the flash crash for the whole trading day. T- Test is implemented on the complete dataset. (ALL OBS) and on size quintiles. ***, ** is the statistical significance at 1% and 5% respectively

AT- Intensity (%)						
Date and Time	All OBS	Q1	Q2	Q3	Q4	Q5
OCT 5th	73.0032	52.8661	65.0270	70.0136	80.6758	90.6372
OCT 4 and OCT 8	73.7079	52.2486	65.7689	70.9584	81.6963	91.5812
Difference in Means	-0.7047***	0.6174**	-0.7419***	-0.9448***	-1.0213***	-0.9439***
Standard Error	(0.119)	(0.289)	(0.262)	(0.248)	(0.195)	(0.115)

Table 3A: T – Test during Flash Crash Day for Algo Trades

In this table, we compare the means of quoted spread and depth of algo trades during the whole trading day of October 5th (Flash Crash Day) with the means of these variables on October 4th and October 8, one day before and after the flash crash for the entire trading day. T- Test is implemented on the complete dataset. (ALL OBS) and on size quintiles. ***, **, * is the statistical significance at 1%, 5% and 10% respectively

Quoted Spread (%) - Algo Trades						
Date and Time	All OBS	Q1	Q2	Q3	Q4	Q5
OCT 5th	1.0907	1.3132	1.2511	1.0985	0.9696	0.8443
OCT 4 and OCT 8	1.0410	1.3168	1.1442	1.0834	0.9428	0.7473
Difference in Means	0.0497***	-0.0036	0.1069***	0.0151***	0.0268***	0.0970***
Standard Error	(0.003)	(0.008)	(0.007)	(0.006)	(0.005)	(0.005)

Depth- Algo Trades						
Date and Time	All OBS	Q1	Q2	Q3	Q4	Q5
OCT 5th	255709	11836.6	72917.5	92263.3	275405	799921
OCT 4 and OCT 8	266331	13537.9	74260.2	103282	354195	748883
Difference in Means	-10622.2***	-1701.3***	-1342.8	-11018.7***	-78790.4***	51038.6***
Standard Error	(2719.4)	(542.7)	(2987.8)	(2561.7)	(6786.5)	(9710.9)

Table 4A: T – Test during Flash Crash Day for Non-Algo Trades

In this table, we compare the means of quoted spread and depth of non-algo trades during the whole trading day of October 5th (Flash Crash Day) with the means of these variables on October 4th and October 8, one day before and after the flash crash for the entire trading day. T- Test is implemented on the complete dataset. (ALL OBS) and on size quintiles. ***, **, * is the statistical significance at 1%, 5% and 10% respectively

Quoted Spread (%) - Non-Algo Trades						
Date and Time	All OBS	Q1	Q2	Q3	Q4	Q5
OCT 5th	2.3090	2.9618	2.5765	2.4099	2.1868	1.8032
OCT 4 and OCT 8	2.3346	3.0999	2.5887	2.4362	2.0184	1.4908
Difference in Means	0.0554***	-0.1381***	-0.0122	-0.0263	0.1684***	0.3124***
Standard Error	(0.012)	(0.034)	(0.029)	(0.029)	(0.026)	(0.018)

Depth- Non-Algo Trades						
Date and Time	All OBS	Q1	Q2	Q3	Q4	Q5
OCT 5th	23304.6	6288	15681.6	22130.8	24529.3	48718.1
OCT 4 and OCT 8	22370	6194.7	15806.8	18613.7	29398.4	42958.6
Difference in Means	934.2***	93.2837	-125.3	3517.1***	-4869.1***	5759.5***
Standard Error	(283.6)	(441.0)	(456.3)	(570)	(781.5)	(924.7)

Figure 1A: Quoted Spread on October 5th, 2012 by size quintiles

The horizontal axis represent time (in hrs) and the vertical axis represents quoted spread in percentage.

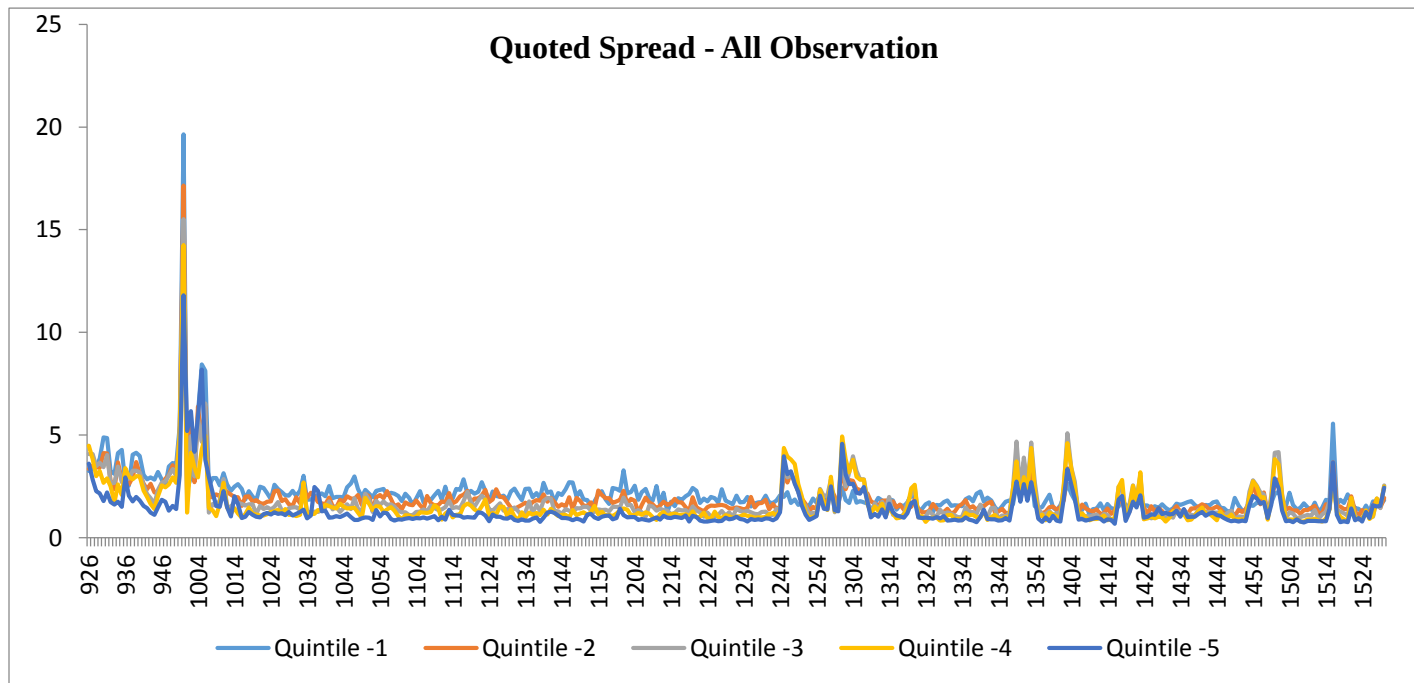


Figure 2A: Depth on October 5th, 2012 by size quintiles

The horizontal axis represent time (in hrs) and the vertical axis represents total depth.

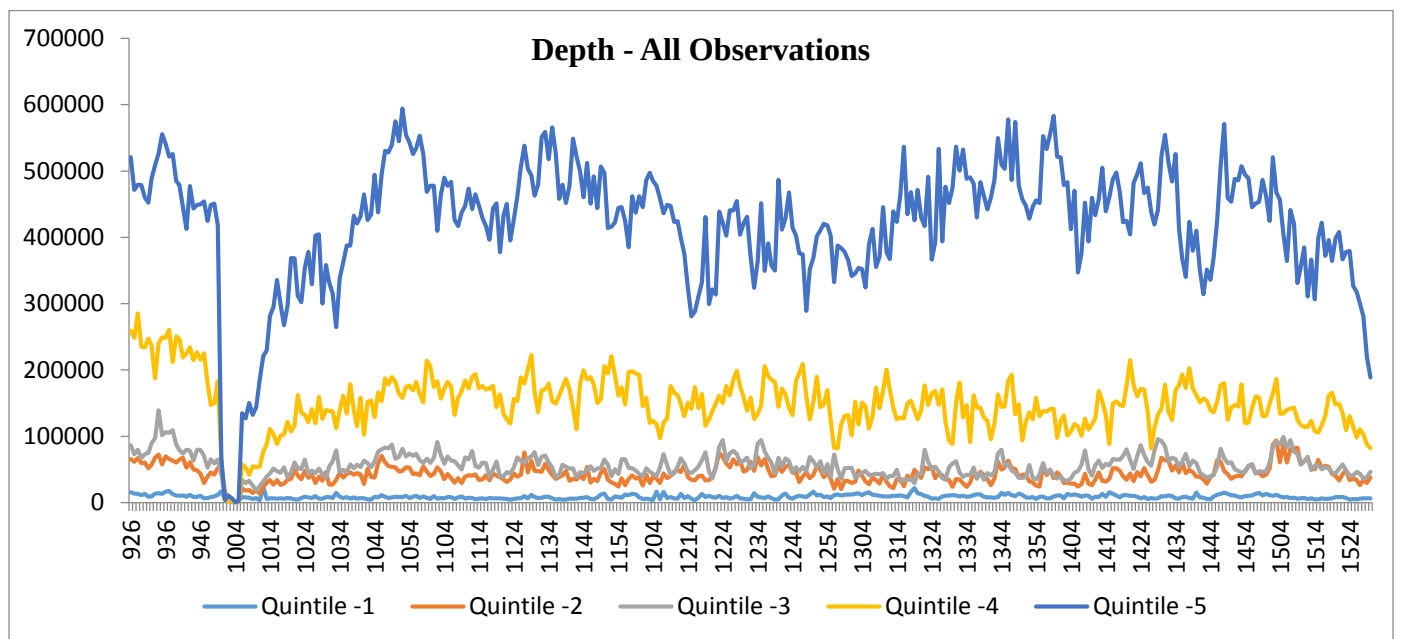


Figure 3A: At-Intensity on October 5th, 2012 by size quintiles (9:25hr – 15:30hr)

The horizontal axis represent time (in hrs) and the vertical axis represents AT-Intensity in percentage.

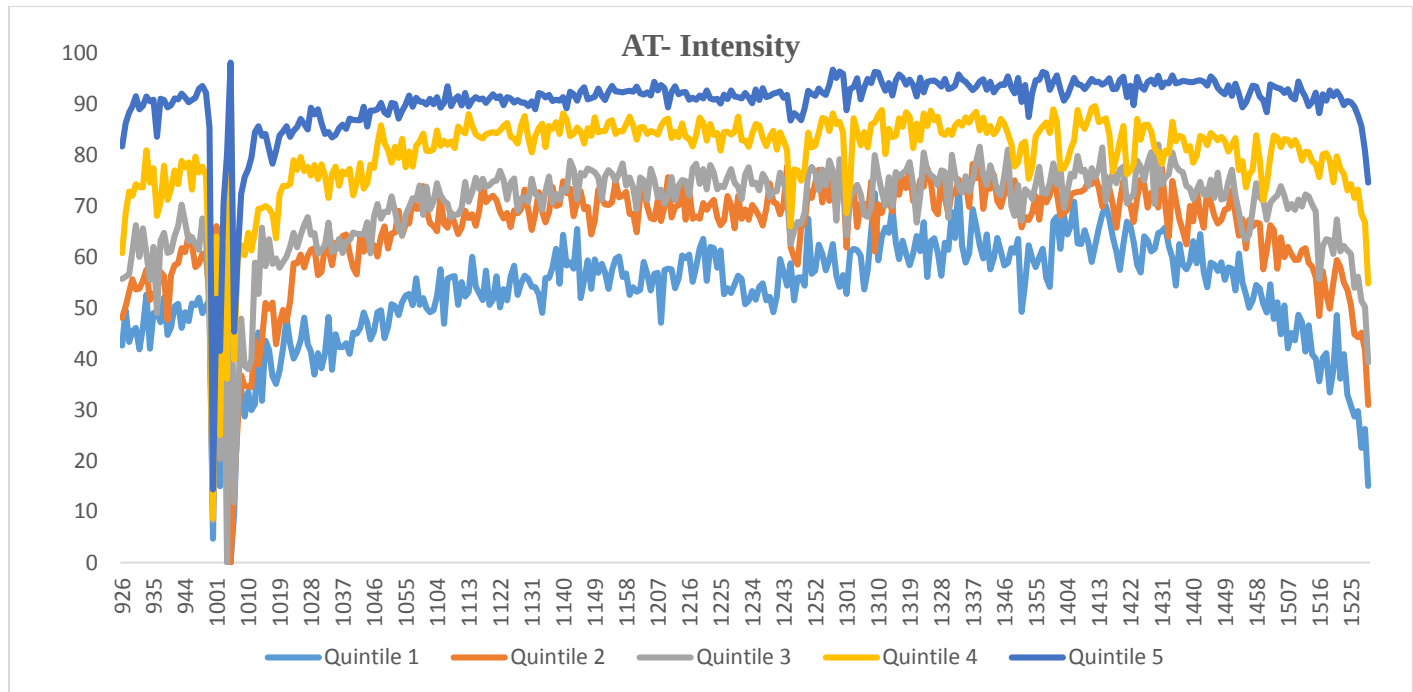


Figure 4A: Quoted Spread on Oct 5th, 2012 by size quintiles for Algo Trades

The horizontal axis represent time (in hrs) and the vertical axis represents quoted spread in percentage.

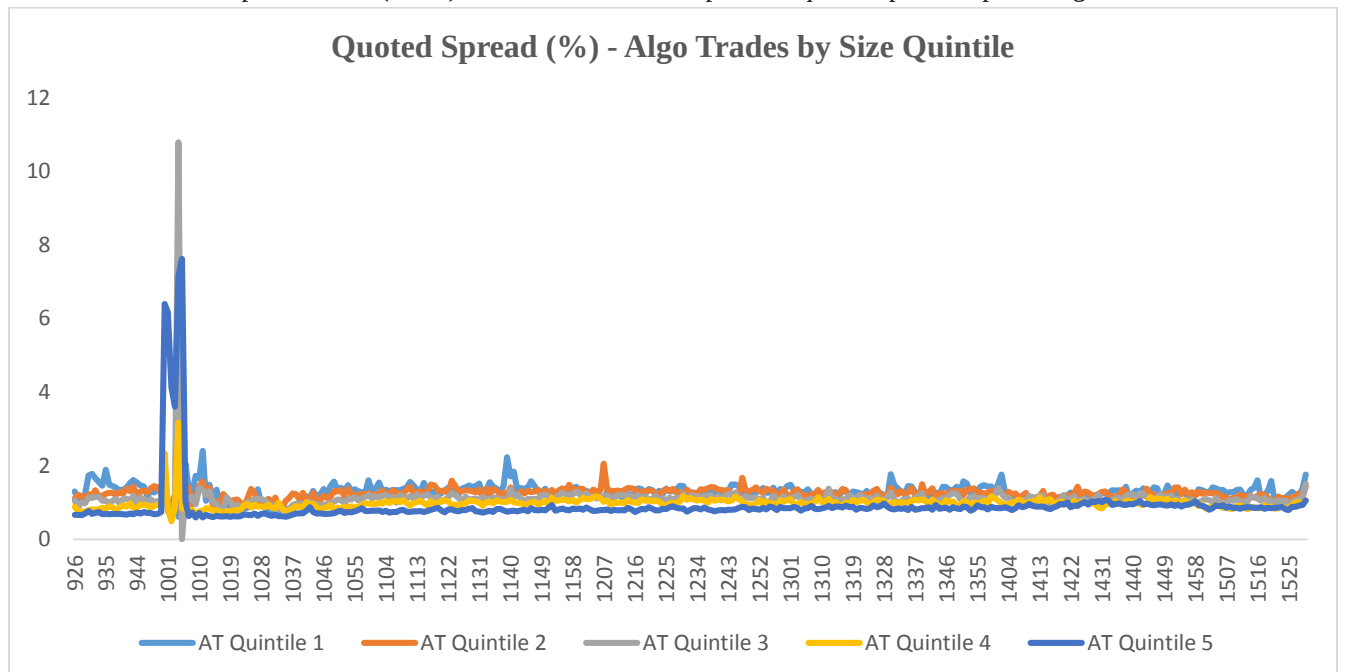


Figure 5A: Depth on October 5th, 2012 by size quintiles for Algo Trades

The horizontal axis represent time (in hrs) and the vertical axis represents quoted spread in percentage.

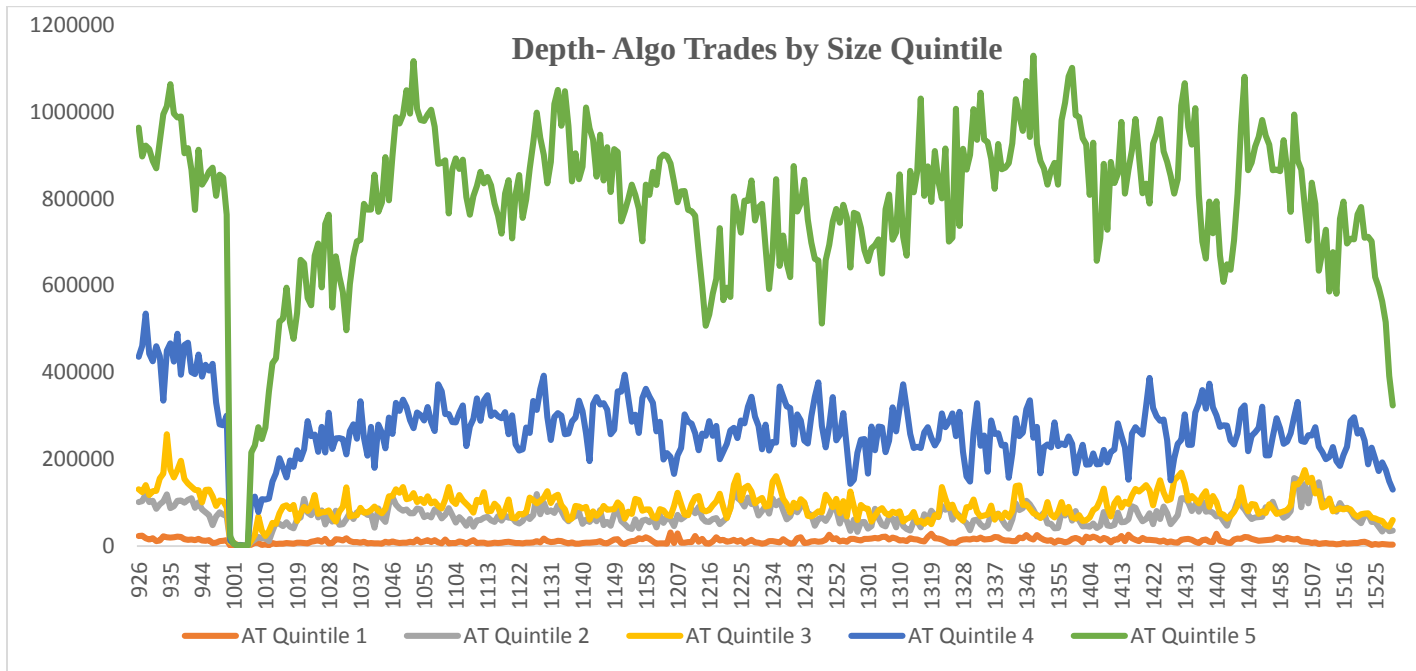


Figure 6A: Quoted Spread on October 5th, 2012 by size quintiles for Non-Algo Trades

The horizontal axis represent time (in hrs) and the vertical axis represents quoted spread in percentage.

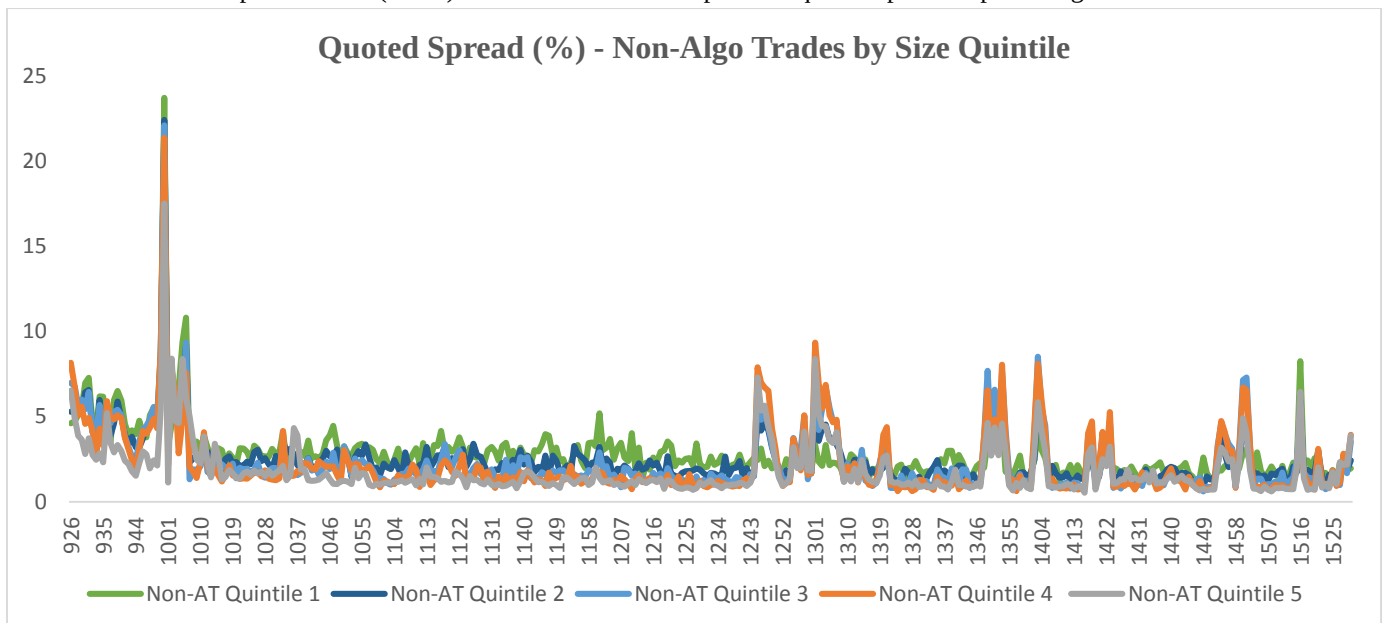


Figure 7A: Depth on October 5th, 2012 by size quintiles for Non-Algo Trades

The horizontal axis represent time (in hrs) and the vertical axis represents quoted spread in percentage.

